In [12]:
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#DATA CLEANING + DATA PREPROCESSING

#IMPORT LIBRARIES
import pandas as pd
import numpy as np

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

#LOAD DATASET
df = pd.read_csv("pbl_dataset.csv")
print("DATASET LOADED SUCCESSFULLY")
print("Rows:", df.shape[0], "Columns:", df.shape[1])

#CHECK MISSING VALUES
print("\nMISSING VALUES BEFORE CLEANING")
print(df.isnull().sum())

#HANDLE MISSING VALUES PROPERLY
# Numerical columns
numerical_columns = df.select_dtypes(include=np.number).columns
for column in numerical_columns:
    df[column] = df[column].fillna(df[column].median())
# Categorical columns
categorical_columns = df.select_dtypes(include='object').columns
for column in categorical_columns:
    df[column] = df[column].fillna(df[column].mode()[0])
print("\nMISSING VALUES AFTER CLEANING")
print(df.isnull().sum())

#REMOVE DUPLICATES
print("\nDUPLICATE RECORDS")
print("Duplicate Rows Found:", df.duplicated().sum())
df = df.drop_duplicates()
print("Duplicate Rows Removed Successfully")

#CLEAN COLUMN NAMES
df.columns = df.columns.str.strip().str.replace(" ", "_")
print("\nCLEANED COLUMN NAMES")
print(df.columns)

#STANDARDIZE CATEGORICAL VALUES
for column in categorical_columns:
    df[column] = df[column].astype(str).str.upper().str.strip()
print("\nTEXT STANDARDIZATION COMPLETED")

#HANDLE NEGATIVE VALUES
for column in numerical_columns:
    df[column] = df[column].apply(lambda x: abs(x) if x < 0 else x)
print("\nNEGATIVE VALUES HANDLED")

# REMOVE OUTLIERS USING IQR
for column in numerical_columns:
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR
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    df = df[(df[column] >= lower_limit) & (df[column] <= upper_limit)]
print("\nOUTLIERS REMOVED")

# FEATURE SCALING
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("\nFEATURE SCALING COMPLETED")

#LABEL ENCODING
label_encoder = LabelEncoder()
for column in categorical_columns:
    df[column] = label_encoder.fit_transform(df[column])
print("\nLABEL ENCODING COMPLETED")

if "Label" in df.columns:
    X = df.drop("Label", axis=1)
    y = df["Label"]
print("\nFEATURES AND TARGET CREATED ")
print("Feature Columns:", X.columns)

#CONVERT SCALED DATA BACK TO DATAFRAME
scaled_df = pd.DataFrame(X_scaled, columns=X.columns)

# Add Label column back
scaled_df["Label"] = y.values


print("\nPREPROCESSED DATASET")
print("Rows:", scaled_df.shape[0], "Columns:", scaled_df.shape[1])

print("\nFIRST 5 ROWS")
print(scaled_df.head())


scaled_df.to_csv("cleaned_dataset.csv", index=False)

print("\nDATA PREPROCESSING COMPLETED SUCCESSFULLY")
print("Final file saved as: cleaned_dataset.csv")
```

```
DATASET LOADED SUCCESSFULLY
Rows: 2000 Columns: 12

MISSING VALUES BEFORE CLEANING
Packet_Size              0
Connection_Duration      0
Error_Rate               0
Protocol                 0
Source_Port              0
Destination_Port         0
Bytes_Sent               0
Bytes_Received           0
Login_Attempts           0
Failed_Logins            0
Encryption_Type        509
Label                    0
dtype: int64

MISSING VALUES AFTER CLEANING
Packet_Size            0
Connection_Duration    0
Error_Rate             0
Protocol               0
Source_Port            0
Destination_Port       0
Bytes_Sent             0
Bytes_Received         0
Login_Attempts         0
Failed_Logins          0
Encryption_Type        0
Label                  0
dtype: int64

DUPLICATE RECORDS
Duplicate Rows Found: 0
Duplicate Rows Removed Successfully

CLEANED COLUMN NAMES
Index(['Packet_Size', 'Connection_Duration', 'Error_Rate', 'Protocol',
       'Source_Port', 'Destination_Port', 'Bytes_Sent', 'Bytes_Received',
       'Login_Attempts', 'Failed_Logins', 'Encryption_Type', 'Label'],
      dtype='object')

TEXT STANDARDIZATION COMPLETED

NEGATIVE VALUES HANDLED

OUTLIERS REMOVED

FEATURE SCALING COMPLETED

LABEL ENCODING COMPLETED

FEATURES AND TARGET CREATED
Feature Columns: Index(['Packet_Size', 'Connection_Duration', 'Error_Rate', 'Prot
ocol',
       'Source_Port', 'Destination_Port', 'Bytes_Sent', 'Bytes_Received',
       'Login_Attempts', 'Failed_Logins', 'Encryption_Type'],
      dtype='object')
```

```
PREPROCESSED DATASET
Rows: 1594 Columns: 12

FIRST 5 ROWS
   Packet_Size  Connection_Duration  Error_Rate  Protocol  Source_Port  \
0    -1.099506             1.415639   -1.734645 -0.331996    -1.164463   
1     0.841882            -1.379689    1.046851 -2.001403     0.355003   
2    -0.120067             1.415639    0.959929  1.337410    -1.661799   
3    -1.076186             1.415639    0.177633 -0.331996     0.192818   
4    -1.280236            -0.680857   -1.517341 -0.331996    -0.615265   

   Destination_Port  Bytes_Sent  Bytes_Received  Login_Attempts  \
0         -0.837451    0.247243       -1.091051       -1.205714   
1         -0.062058   -0.924326       -0.289630        0.754417   
2         -0.837451   -0.793425       -1.150051        0.509401   
3         -0.719967    0.134013       -1.417737       -1.205714   
4          2.052647   -1.321395       -1.036284       -1.205714   

   Failed_Logins  Encryption_Type  Label  
0      -0.872496         1.520256      0  
1      -0.175549        -0.893378      1  
2      -0.175549         1.520256      1  
3      -0.872496         1.520256      0  
4      -0.872496         1.520256      0  

DATA PREPROCESSING COMPLETED SUCCESSFULLY
Final file saved as: cleaned_dataset.csv
```

In [ ]: